

SCHOOL LIBRARY MANAGEMENT SYSTEM

Case Study

Computer-based Library Management system

The library of Kiongwani Secondary School has 3,000 textbooks. Each book is identified by its *Author*, International Serial Book Number (*ISBN*), *Book ID* and *Title*. The books are arranged on the shelves using their book IDs and Card catalogues are maintained for all the books. There are two types of card catalogues; one is arranged according to the author's names, while the other is arranged according to the Titles of the books.

Each student is issued with three borrower cards that have a Registration number and Name of the student. To locate a book for borrowing, a student checks in the card catalogue for its classification, then moves to the shelf to retrieve it. The student surrenders a borrower's card at the issuing counter where the librarian gives out the book and stamps the date of return.

A student is not allowed to borrow more than three books at any one time. Students are charged for overdue books at a fixed rate multiplied by the number of days

Problem analysis

The system analyst is required to:

- i). Identify & understand the problem.
- ii). Find out much about the existing system (whether manual or computerised) in order to come up with a good & relevant proposal for the new system.

From the School library management system, the problem is to replace the inefficient manual card cataloguing operations with an efficient computerized system.

The system analyst must try to answer the following questions:

1. What are the shortcomings of the current system?
2. What types of records are used for books and students in the library?
3. What procedure is followed to borrow or lend books?
4. How are overdue books handled when returned?

Information gathering

If the management responds and gives the system development team a go ahead to start developing the information, the members have a task to discover user requirements of the new system through collection of facts.

Interviews

Below is a verbatim introduction of sample interview with the library manager.

INTERVIEW TITLE	
BRIEF INTRODUCTION	Interviewer: Interviewee:
Interviewer: Hello	
Interviewee: Hello. Welcome to my office.	
Interviewer: Thank you. Please call me John. I would like to ask you a few questions about the system that we are developing.	
Interviewee:	

Questionnaires

Below is an extract of a questionnaire used to gather data from library attendants.

QUESTIONNAIRE	
BRIEF INTRODUCTION	Date:
.....	
.....	
QUESTIONS	
1. How long have you worked as a library attendant?	
☐ 1 yr.	☐ 2 yrs.
☐ Over 2 yrs.	
2. How long does it take to rearrange books on the shelves?	
☐ Days.	☐ Weeks.
☐ Months.	

Preliminary investigation report:

The *preliminary investigation report* dwells on the problems of the existing system from the general point of view.

Fact-finding report:

Fact-finding report gives a more detailed analysis of the existing system and proposals for improving or overhauling it altogether.

The figure below shows a sample general outline of the fact-finding report presented to the management of the school library and the head of the school.

Library Management information system

Fact-finding report

1.0 Table of contents.

2.0 Executive summary:

2.1 Objectives: The new computerized system is intended to improve efficiency in the library by:

- a). Keeping an inventory of all the books in the library and automatically updating the stock, hence eliminating the tedious physical counting process.
- b). Reducing the time needed to seek for a book by 60%.
- c). Tracking overdue and lost books.

2.2 Recommendation:

This system could result in efficient processing of library transactions. It will replace the tedious manual system.

3.0 Methods used to study the system.

3.1 Interviews: - were used when seeking facts from management.

3.2 Questionnaires: - were circulated to user staff.

3.3 Observation: - I observed as books were searched and issued.

4.0 Detailed results:

4.1 Problems: - duplication of records, delays and book losses.

4.2 Opportunities: - improve efficiency, improve stock management, etc.

4.3 Alternatives: - enforce controls in the current system, employ more staff, etc

5.0 Summary:

The new system is highly recommended because the other alternative of enforcing controls and employing more staff will add operating costs with little additional value.

Requirements specifications

The following requirements specifications are considered:

1. Output specification.
2. Input specification.
3. File/data structures.
4. Hardware & software requirements.

Output requirements.

In our case study, the outputs required from the Library management system include:

1. A report about all the overdue books showing charges against each borrower.
2. A search report for a particular book showing its classification and whether it's on the shelves or not.
3. A search report about a particular student showing which books he/she is currently holding.

The table below shows a sample report expected to be generated from the computerized library system showing all the overdue books.

OVERDUE BOOKS					
ISBN Number	Title	Student Name	Date Issued	Date Due	Overdue Days
1. 966 49 574	Computer Studies	Samsom Tito	10/01/2005	24/03/2005	6
2. 966 49 600	Mathematics Form 2	Linda Kiamba	12/01/2005	26/03/2005	4
3.					
4.					

Input specifications.

The system analyst should identify the input needed to obtain the relevant information from the system.

In our case study, the following inputs can be deduced from the output specification:

1. The type of data needed to add a book to the Books file/database in the library,

For example;

In the Library database, the following data items may be entered:

- a). Title of the book.
- b). Names of the author(s) of the books.
- c). The ISBN number of the book.
- d). Book ID.

2. Determine the data needed for someone who wishes to borrow a book.

After identifying all the inputs, the analyst designs the user interface by designing data entry forms or screens.

The figure below shows a form used for the registration of a new member:

New member registration	
Registration Number:	
First name:	
Surname:	
Gender:	
Save Exit	

NB:

- J Textboxes& Command buttons placed on the forms must be neatly aligned & balanced on the form.
- J The size of the form should not be too small or too big.
- J The interface should not have too bright colours.

System Design

In the design stage, the analyst may use a system flowchart to solve the problem at hand.

Designing a system flowchart.

A system flowchart gives a summary of how particular processes are done within the business organization. It allows one to break a process down into individual events/activities, and also display these events in a short form showing the sequential or logical relationships between them.

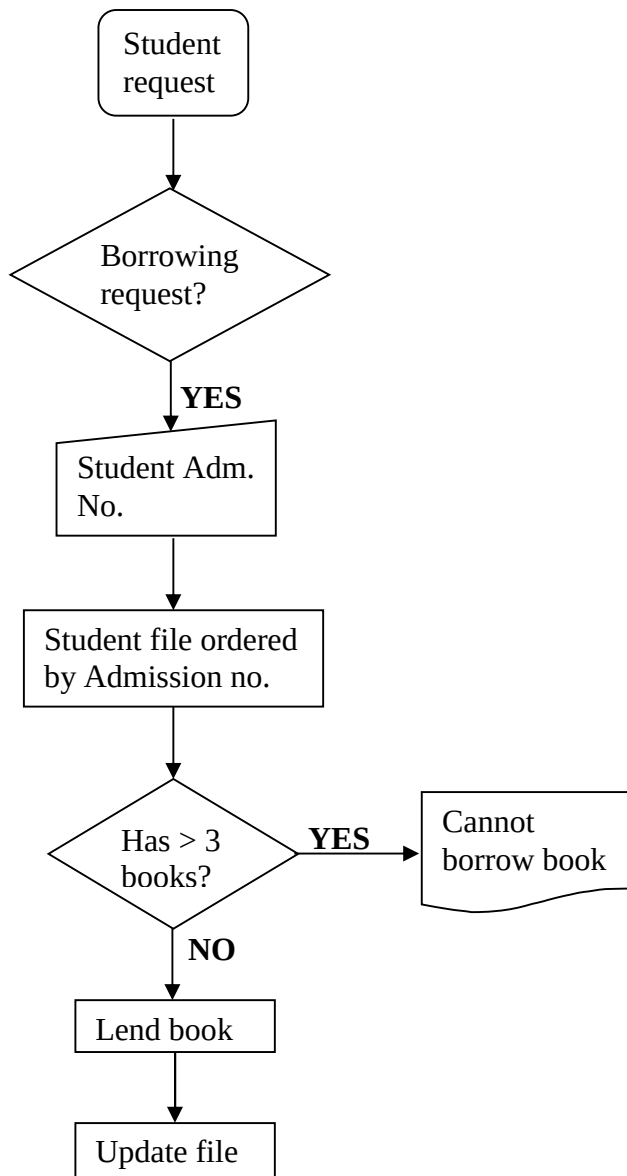
Important guidelines when designing a system flowchart:

1. Start by writing the title of the flowchart, e.g., **Library Books Management Information System**.

2. Start drawing the flowchart with the trigger event, e.g., our trigger event would be *a student request to borrow a book or return an overdue book*.
3. List down the actions taken in their logical order until the process is concluded. Use few words to describe the actions.
4. In case there are many alternatives at the decision stage, follow the most important and continue with it. The less important alternatives can be drawn elsewhere and reference made to them using the On-page or Off-page connectors.

A books borrowing system flowchart:

The figure below shows a sample books borrowing system flowchart for the proposed computerised library management system for the school.



Explanation

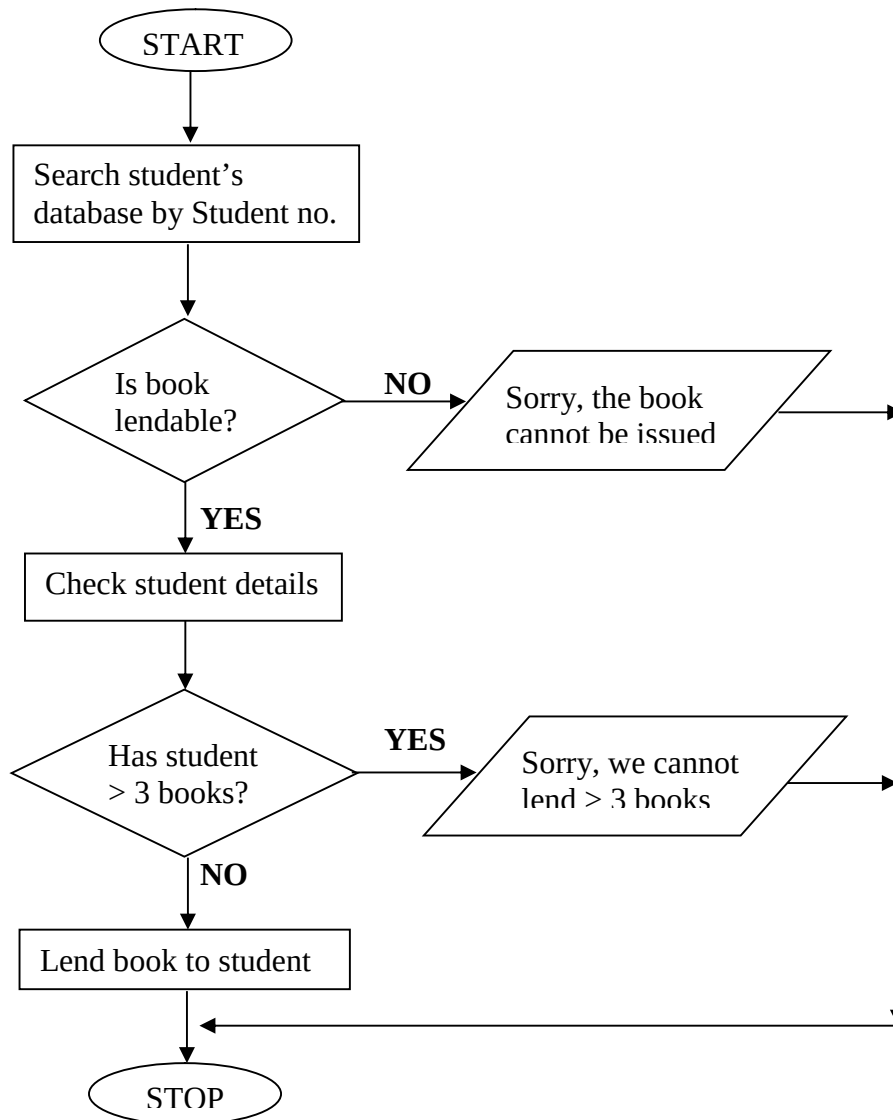
From the system flowchart, we observe that;

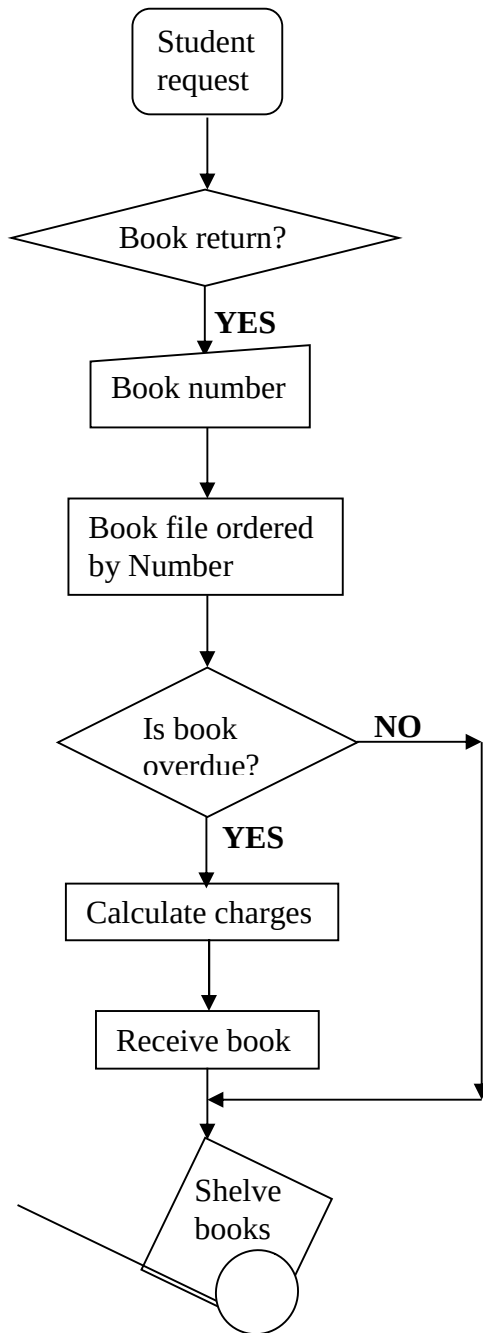
1. A student requests for a particular book.
2. The system checks for the student's record in the system. If the student has more than three (3) books, a message is displayed and the student cannot borrow an extra book.
3. If the student has less than 3 books, then the book can be given out to him/her, and the student file updated.

NB: A system flowchart can be used to extract a program flowchart for a particular task.

A book lending/borrowing program flowchart:

The figure below illustrates a book lending/borrowing program flowchart that has been extracted from the Library management system flowchart.



A book return system flowchart:**Practical exercise.**

Extract the book return module flowchart from the above system flowchart.

System construction

A database package can be customized to construct a system.

Below is a sample data entry form constructed to enable entering a new book record into the Library information database.

Books: form	
Book Number:	
ISBN Number:	
Publisher Name:	
Year Published:	
Title:	
Purchase Price:	
Date Purchased:	

Testing the system

E.g., if one of the requirements of the computerised library management system is to ensure that no member is allowed to borrow more than 3 books at the same time, the system must be able to display a message box to that effect without fail.

Sample test data

To test whether the new computerized information system is working as expected, test data is used for every module (subsystem).

For example;

In our library case study, we need to test sample data for books entry, books borrowing, etc. The table below shows a sample test data that can be entered in the database whenever a book is borrowed.

Last name	Middle name	Class	Stream	Book borrowed	Date borrowed	Return date
Mburu	James	3	N	Maths Form 1	8/6/2006	26/6/2006
Janet	Achieng	3	S	Kamusi sanifu	23/6/2006	30/6/2006
Helen	Mutua	4	E	Elementary Chemistry	6/6/2006	24/6/2006
Ali	Mohammed	2	W	Computer Studies Form 2	6/5/2006	23/5/2006
Kerich	Dennis	1	X	Biology Form 1	6/6/2006	24/6/2006
James	Kamau	3	S	Computer Studies Form 3	7/6/2006	25/6/2006

NB: In real situation, you are expected to provide as many test data as possible to prove the functionality of your system.

Output reports

To prove that the system is working and giving the desired result, you should provide a number of sample output from various system modules.

Below is a sample report showing a list of members who have borrowed books.

Kiongwani Secondary School Books Borrowed Report						
Last Name	Middle Name	Class	Stream	Book Borrowed	Date Borrowed	Return Date
Mutinda	Gianmaria	3	S	Maths Form 1	8/1/2007	3/4/2007
Muia	Veronica	3	N	Kamusi Sanifu	23/2/2007	29/3/2007
Musyoka	Cyrus	4	N	Elementary Chemistry	6/2/2007	24/3/2007
Muindi	Stephen	2	S	Computer Studies	6/1/2007	23/3/2007
Ahinoab	Miriam	1	N	Biology Form 1	6/2/2007	24/3/2007
Njoroge	Martin	3	S	Computer Studies	7/1/2007	25/3/2007
Tarayia	Josephine	3	S	English Form 3	7/1/2007	25/3/2007
Mutua	Stelamarris	2	N	Computer Studies	8/2/2007	2/4/2007

Files and data stores design

There are two methods of data storage in a computer: *Flat files* and *databases*.

Flat files

In flat files, records are stored in separate files each with particular details.

The use of separate files means that; the same data is usually stored in more than one place.

Database systems

A *database* is a centrally controlled store of data. It allows different users who share the database to have access to different data.

Note. Storing of data using a database system eliminates the validation problems of the flat-file system.

For the Library system, we will need to store data about books and students as entities of the system in a database.

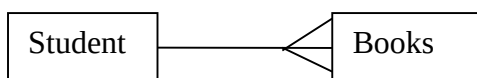
There is also need to define a relationship for these entities during the database design.

Entity relationship (E-R) diagrams

ERD is a diagram that shows the logical relationship between data entities.

For example,

A student can borrow many books. This can be represented in the database design as a “*one-to-many relationship*” as shown below.



The line that joins the two entities divides into three on the Books side to show that one student can borrow many books.

Other relationships that can be created between entities are “*one-to-one*” and “*one-to-many*”.

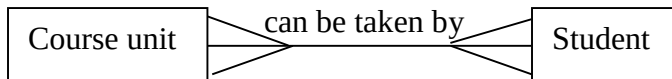
One-to-one relationship:

e.g., one student can only have one registration number.



Many-to-many relationship:

e.g., in a school environment, one course unit can be done by many students, and one student can do many course units.



NB: It is possible to have a database design that has all the three entity relationships. However, when designing your database, you should develop One-to-many relationships between entities.

Defining attributes

You need to identify the attributes for each of the entities in the E-R diagram. The attributes will become the **fields** of the database table that will store data for each entity. A complete set of attributes in a table will form a *record*.

Each entity will have one attribute that will hold a unique value for each of the records entered in the database. This unique attribute is called the **Primary key** (or, **key attribute**).

To create a relationship between two tables, create a field in the second table that has the same data type as the primary key of the first table.

This field created in the second table is called the **foreign key**.

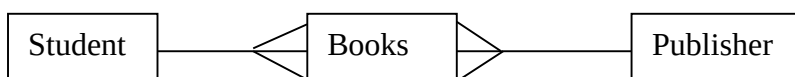
Entities and their attributes	
Entity	Attribute
Book	Book Number (<i>primary key</i>) ISBN number Title Year Published Purchase price Registration number (<i>foreign key</i>) Publisher ID (<i>foreign key</i>)
Student	Registration number (<i>primary key</i>) First name Middle name Class Stream
Publisher	Publisher ID (<i>primary key</i>) Publisher name Address City Phone number

NB: Any attributes that will encourage duplicate storage of data in the database, should be separated off and used to form their own tables.

E.g., one “Publisher” may have published many books. Hence, if you have one hundred different books in the library done by the same publisher, you will have to enter the same name a hundred times in the database.

The process of trying to eliminate storage of duplicate values in the database is called *normalization*.

The E-R diagram of the system will be as shown below:



This E-R diagram shows that a student can borrow many books. Likewise, a publisher can publish many books.

Creating the database.

To create a database,

1. Click on **Start**, then select **Microsoft Access**.
2. From the dialog box that appears, select the **Blank Access Database** option, then click the **OK** button.
3. In the *File New Database* dialog box, type the name of the new database to be created in the *File name* box.
4. In the *Save in* box, select the folder where you want to save your database, then click the **Create Database** button to display the Database window, which will allow you to create the various tables.

Defining a table/file structure.

To define the table/file structure;

1. Click the **Tables** tab in the database window, then double-click the **Create table in Design view** command.
2. Create the three tables, i.e., students, books, and publisher using the attributes you had defined.

Setting the primary key and foreign key

A *Primary key* is a unique field that identifies each record in a database table.

This means that, a primary key;

- (a) Holds unique values for each record.
- (b) Cannot accept a duplicate, e.g., if one student has the registration number 2000, no other student can be assigned the same number.
- (c) Cannot accept null or empty values, i.e., a value must be entered into the field before proceeding to another record.

The *Foreign key* is included as an attribute of an entity in order to create a relationship with another entity in the database.

E.g., the *registration number* field in the BOOKS entity creates a relationship between the STUDENT entity and the BOOKS entity.

- (a) The entity with the foreign key is usually on the many side of a relationship.
- (b) Values in a foreign key field of a table in a database are unique but are allowed to repeat as long as they appear in the primary key of the primary table.

To define table relationships;

1. Click on **Tools**, then click **Relationships**.
2. From the *Show table* dialog box that appears, select the tables, and click **Add** so that they are displayed on the relationship window.
3. Drag the primary key of the first table to the foreign key of the second table.

For example;

To create a relationship between STUDENTS table and BOOKS table, drag the *registration number* of the Students table to the *registration number* of the Books table.

4. Enforce referential integrity: *Referential integrity* governs the nature of records in a “one-to-many relationship”.

This means that, all foreign keys in the child table must have a matching record in the parent table. Therefore, you cannot add a record in the child table if it does not have a related record in the parent table.

Click to select “*Cascade Update Related Fields*” if you want your database to update the child table once the parent record is updated.

This means that, if the student's registration number is changed, all the records in the related tables will automatically be updated.

Data entry into the system.

Once you have created your database tables and their relationships, you can then enter data into the tables using forms which you create.

Fig. 1: The Students' records form can be used by the library attendant to enter students into the database.

Students' records: form	
Registration Number:	1353
First Name:	Veronica
Middle Name:	Mwikali
Class:	3
Stream:	N

Fig. 2: The Books' details form can be used to enter books into the database.

Books' details: form	
Book Number:	0001
ISBN Number:	891001
Publisher Name:	Longhorn Publishers
Year Published:	2005
Title	Computer Form 3
Purchase Price:	Ksh. 210.00
Student Number:	2475
Print Save Close	

Fig. 3: The Publishers' entry form can be used to enter publisher records into the database.

Publishers' entry: form	
Publisher ID:	K003
Publisher Name:	Longhorn
Address:	Box 020-34769
City:	Nairobi
Phone Number:	020-345627

Reports

Below is a sample report generated from the database showing a summary report indicating the books owed by students.

The report has been generated from a query that is created using the three related tables.

Books owed to the Library					
<i>Class</i>	<i>First Name</i>	<i>Middle Name</i>	<i>Stream</i>	<i>Title</i>	<i>Publisher</i>
<i>Form 2</i>					
	Ole Lisa	Peter	S	Computer F2	Longhorn
	Ali	Mohammed	N	Computer F2	Longhorn
<i>Form 3</i>					
	Kerich	Judith	S	Advanced Maths	KIE
	Atieno	Achieng	N	Kamusi Sanifu	Bin Salim
	Mburu	James	N	F3 English	Ndege Books
<i>Form 4</i>					
	Helen	Mutua	S	Computer F4	Longhorn

Automating the database

After creating the basic database objects, i.e., tables, forms, reports and queries, you may wish to automate operations such as data entry, printing of reports, or make the user interface more user friendly.

Some of the tools used to automate a database in Ms-Access are:

(a) Switchboard (main form):

Switchboard is a special form used to automatically access other database objects such as forms & reports.

The switchboard is the user-interface (or main menu) that is loaded when an application is started. The form mainly contains command buttons that enable the user to select the task to work on.

To create a switchboard;

1. Click on **Tools**, point to **Database Utilities**, then click **Switchboard manager**.
2. If Ms-Access prompts you whether you want to create a switchboard, click **Yes**.
3. From the *Switchboard manager* dialog box, click **Edit**.
4. From the *Edit switchboard Page* dialog box, type the name of the switchboard in the *Switchboard Name* box, e.g., *Main Switchboard*. If there are no menu items on the switchboard, then click **New**. To edit the menu items, click **Edit** to display the *Edit*

Switchboard Item dialog box in which you can specify the menu items to be included on the switchboard.

5. In the *Text:* box, enter the name of the menu, e.g., *Students Data Entry*.
From the *Command:* list box, select the object to be accessed, e.g., *Open form in Edit menu*.
In the *Form:* list box, select the name of the object, e.g., *Students' record*, then click **OK**.
6. Finally, click the **Close** buttons to close the switchboard dialog boxes.

(b) **Macros:**

This is a set of one or more actions used to automate commonly performed tasks such as opening & closing forms, and printing reports.

To create a macro in Ms-Access;

1. On the Database window, click the **Macros** tab, then choose **New**. The macros design grid appears.
2. In the *Action* window, click the down arrow to select an action that you want to be executed directly every time you run the macro, e.g., scroll downwards and click *OpenForm*.
3. In the lower part of the macro design window, specify the action arguments.
For example, for the action *OpenForm*, select the name of the form, e.g., *Switchboard, view mode*.

Database security

Ms-Access provides the following features to protect your information system from unauthorized access.

1. Use of Passwords
2. User and group permissions.
3. User-level security.
4. Data encryption.

Password protection

A password is set for any user who wishes to open or edit the database file.

To setup a database password in Ms-Access;

1. Click on **File**, then click **Open**.
2. Locate the file you want to open, then click its filename.
3. Click the arrow next to the *Open* button, then choose **Open Exclusive**.
4. Click on **Tools**, point to **Security**, then click **Set password**.
5. Enter the password in the *password* text box, and re-enter the same password in the *verify* box.
6. Click **OK**.

When a user opens the database, he/she will be prompted to enter the password.

User and group permissions

Operating systems like Windows XP provide the User accounts dialog box that allows an administrator to assign permission to users on database objects such as queries, forms, and reports.

This means that, you must be a member of the Administrators group so that you can be allowed to open, run, or modify the object.

User-level security

This feature is used to assign permission to users in order to limit access to certain database objects.

Data encryption

Encryption is applied if the database is to be used in a network or sent electronically. Only the intended recipient can use the decryption key to read the message.

Setting the Start up options

Start up options enable you customize a database in order to make it more presentable to the users.

Once the startup options have been set, the database will now be starting as an application instead of loading it from Ms-Access application window.

To setup the Start up options for your application;

1. Click on **Tools**, click **Start Up**, then specify the following:
 - ◆ *Applications Title*: - type the title that will appear on the title bar (instead of the default Microsoft Access title), e.g., *Bukuma Distributors Information System*.
 - ◆ *Applications icon*: - choose an icon that would go together with your application title.
 - ◆ *Menu Bar*: - specify the menu options that should appear when running the application.
 - ◆ *Display Form/Page*: - choose the form you want automatically displayed once you start the application, e.g., *Main Switchboard*
2. Ensure the “*Display Database Window*” checkbox is not selected in order to limit the users from viewing the objects created in your database.
3. Click **OK**.

User manual

User manuals are used to help a person use the system with little or no guidance.

The manual must contain information such as:

1. How to install, start and run the system.
2. How the system appears when running (interface).
3. How to carry out various tasks, e.g., how to include a new entry, data entry, how to modify a record, etc
4. A troubleshooting guide, which describes error correction & how to get help when faced with problems.